REVOFINANCE

Spending Behavior & Budget Risk Analysis

Understanding user spending patterns to proactively manage budget risk and financial health

Fitriyatus Sholihah
16-01-2025
RevoU FSDA batch Oct 2025
https://www.linkedin.com/in/fitriyatus-sho/

Executive Summary

Total Expense
Rp509M

Total Transaction
1.000

Avg per Transaction
Rp509K

Key Takeaways:

- Total annual spending reached Rp509M across 1,000 transactions, reflecting high financial activity.
- Overall spending exceeded the planned budget, indicating emerging financial risk.
- Spending risk is not evenly distributed over time; it concentrates in specific months and categories.
- Advanced behavioral clustering reveals that high-risk spending does not always occur during the highest spending months.

Business Background

RevoFinance is a fictional fintech company that empowers individuals in Indonesia to track and manage their personal expenses across both offline and online merchants. With increased financial literacy and mobile app adoption, RevoFinance aims to help user better understand their spending behaviour, optimize personal budget, and take control of their financial health.

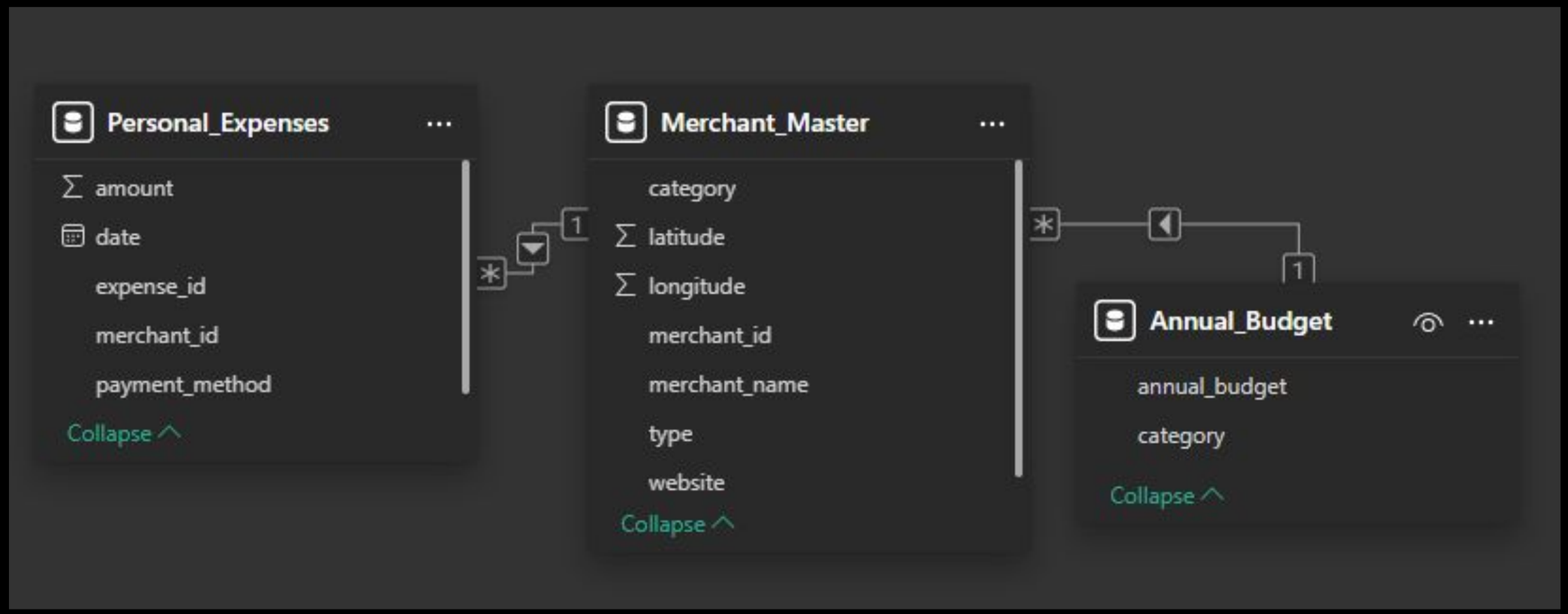
Problem Statement

How to help user to tracking their spending pattern and monitoring their spending versus budget by identifying their overspending and suggesting user where to cut off?

Objective

1. Enable user to track monthly and annual spending trends
2. Visualize spending distribution across categories, locations, and merchants
3. Compare actual spend against budget
4. Provide actionable insight to improve personal financial planning

ERD



Disclaimer:

-This analysis is for educational purposes and does not reflect the actual business metrics of RevoFinance

-Insight and recommendation are derived from provided dataset and may not represent real market conditions

-All insight and dashboard, and metrics should be generated based on personal interpretation and visualization skill

In tableau, to join the dataset I only using relationship, not left join because I don't want to merge the dataset. Cardinality between Merchant and personal expenses is one to many, and between merchant to annual budget is many to one.

Dashboard

REVOFINANCE

Spending Dashboard | Executive

Total Expense	Total Transaction	Avg per Transaction
Rp509M	1,000	Rp509K

Month of Date

Januar Decem

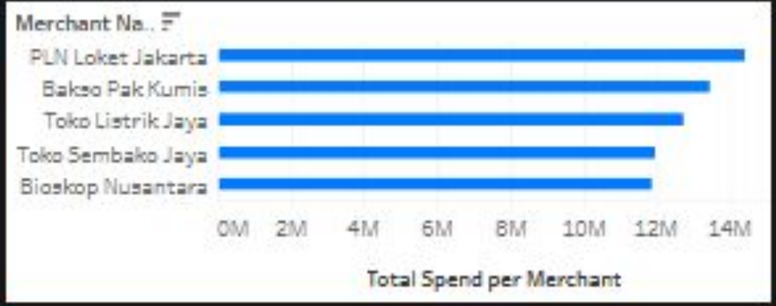
Category

(All)

Type

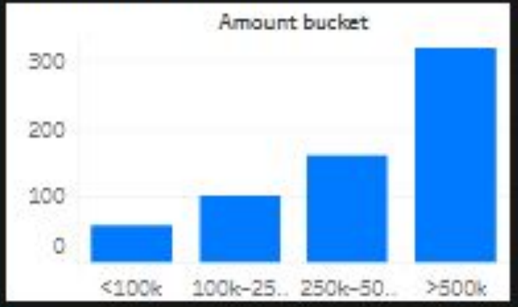
(All)

Spending Concentration: Top 5 Merchants



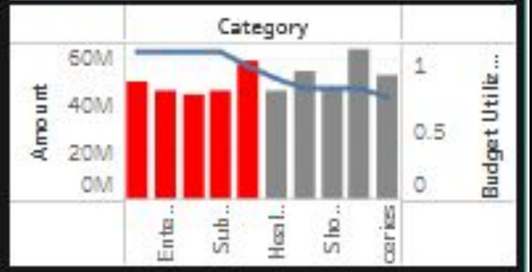
Cummulative Budget Utilization Over Time

Transaction Distribution



Majority of Spend Driven by Few Categories

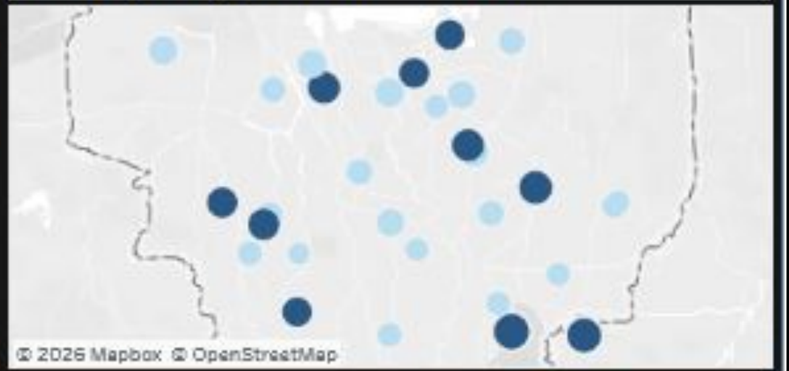
Spending Category vs Budget Utilization



Budget Implementation

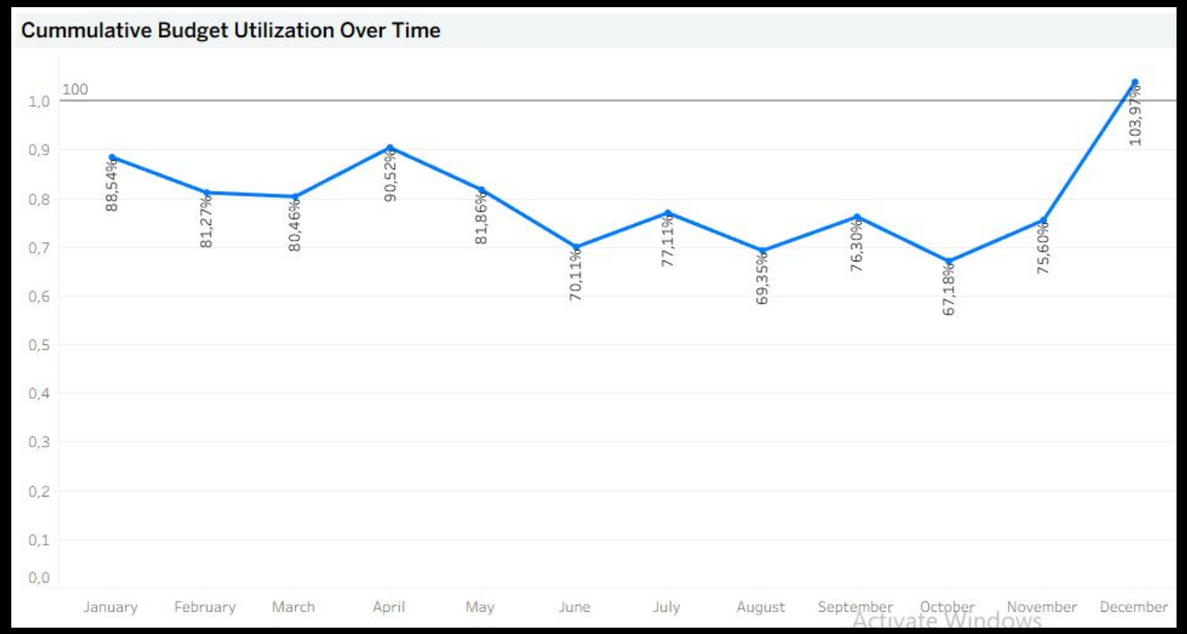
Month, Year of ..	# of Purchase	Amount	Difference in ..	Budget Varia..
January 2024	90	Rp47M		-Rp2M
February 2024	84	Rp41M	-Rp6M	Rp4M
March 2024	85	Rp45M	Rp4M	Rp0M
April 2024	101	Rp47M	Rp1M	-Rp1M
May 2024	87	Rp43M	-Rp3M	Rp2M
June 2024	70	Rp38M	-Rp5M	Rp7M
July 2024	78	Rp41M	Rp2M	Rp5M
August 2024	67	Rp36M	-Rp4M	Rp9M
September 2024	80	Rp40M	Rp4M	Rp5M
October 2024	73	Rp36M	-Rp4M	Rp10M
November 2024	82	Rp40M	Rp5M	Rp5M
December 2024	103	Rp54M	Rp14M	Rp9M

Offline Spending Concentration – Jakarta



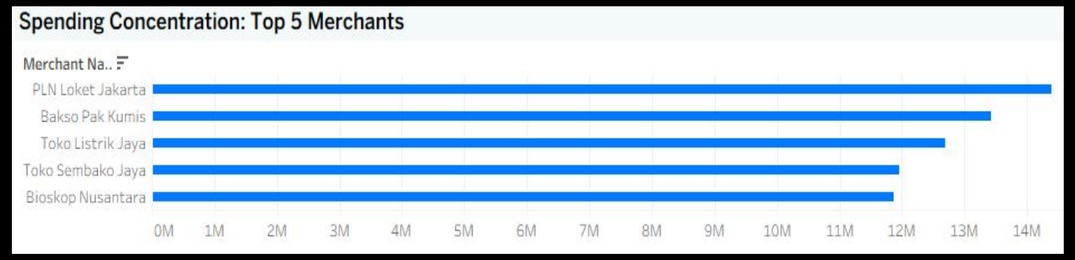
User spending exceeds budget, driven by behavioral rather than structural factors

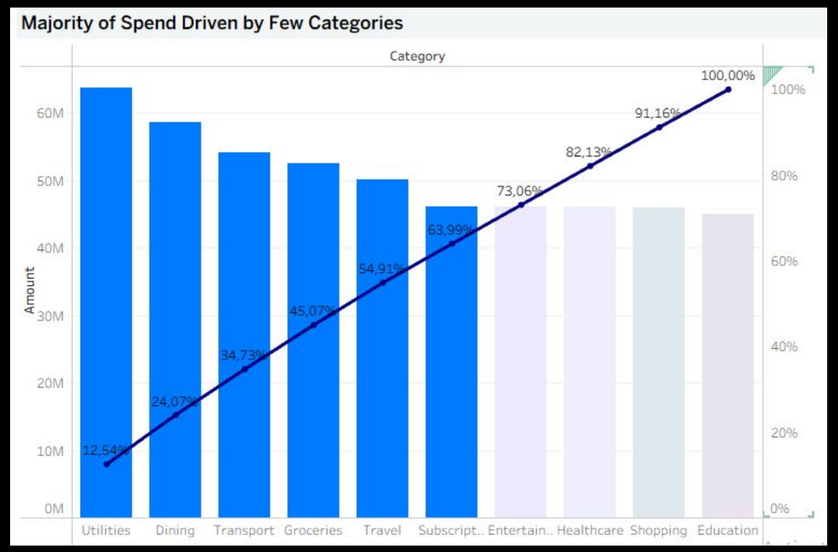
- Average spending intensity is Rp509K per transaction, indicating frequent mid-to-large expenses.
- Cumulative budget utilization crossed 100% toward year-end, confirming budget pressure.
- The budget breach is preventable, as risk builds gradually over time.



Spending is concentrated across a limited number of merchants and categories

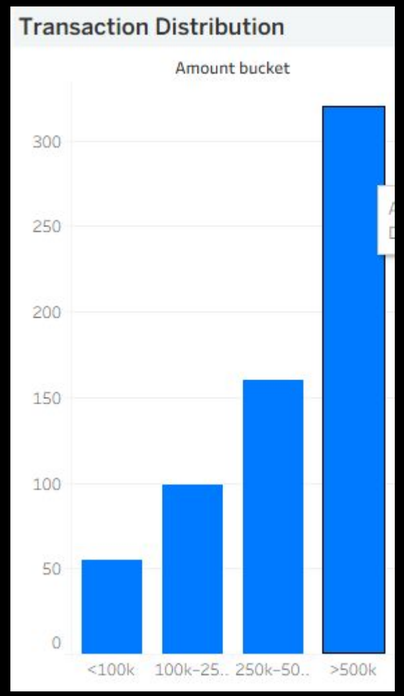
- The **Top 5 merchants** account for a disproportionate share of total spending.
- Pareto analysis shows that a small subset of categories drives the majority of expenses.
- This concentration creates both **risk exposure** and **intervention opportunities**.





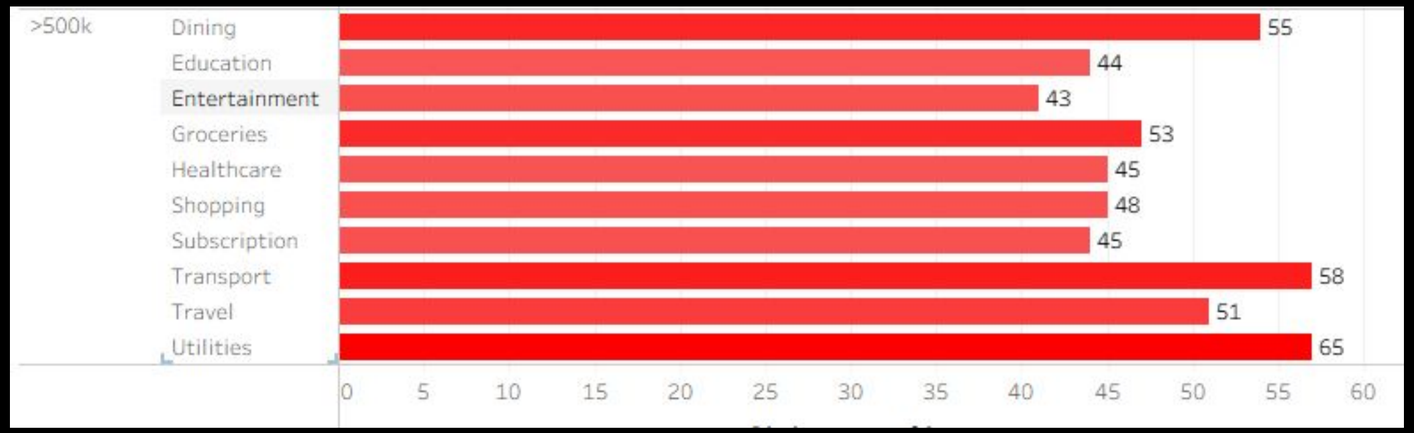
Large transactions are the primary driver of budget risk

- The majority of spending comes from transactions above Rp500K.
- Smaller, daily transactions contribute marginally to overall risk.
- Budget overruns are therefore driven by large discretionary decisions, not routine spending.



High-Frequency "Habitual" Spending in Utilities & Dining

- Utilities emerged as the largest spending category in 2024, totaling Rp63M across 120 transactions, with 54.1% of spending coming from transactions above Rp500K, indicating exposure to recurring high-value expenses.
- Dining has frequency (55x) that matches Necessity (Groceries 53x), showing a high reliance on lifestyle services.
- Across almost all categories, the >500k bucket dominates total spending.
- Travel & Entertainment show mixed behavior, Significant contribution from 250k–500k and >500k, these categories represent discretionary but high-impact spending.

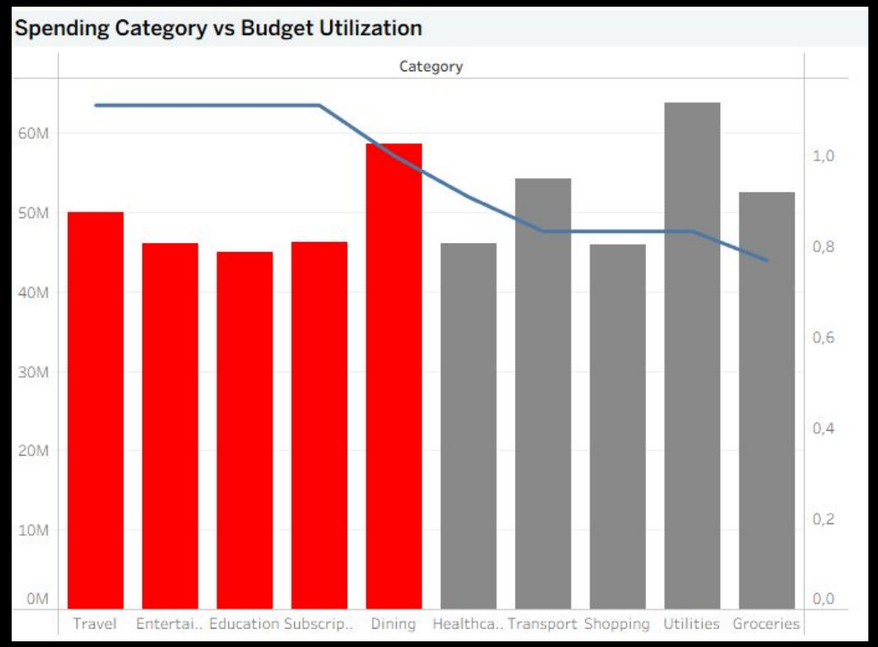


11% Budget Leakage in Lifestyle Categories

Users have exceeded their annual limits by 11.11% in non-essential categories :

- Education
- Entertainment
- Subscription
- Travel

Signaling systemic overspending rather than isolated behavior.



December Surge: 20% Wallet Share Captured by Entertainment

December 2024 represents the highest spending month, with total spend Rp14M higher than November, driven primarily by Entertainment, which alone accounts for 20.23% of user monthly spending in Rp54M.

Budget Utilized per Month

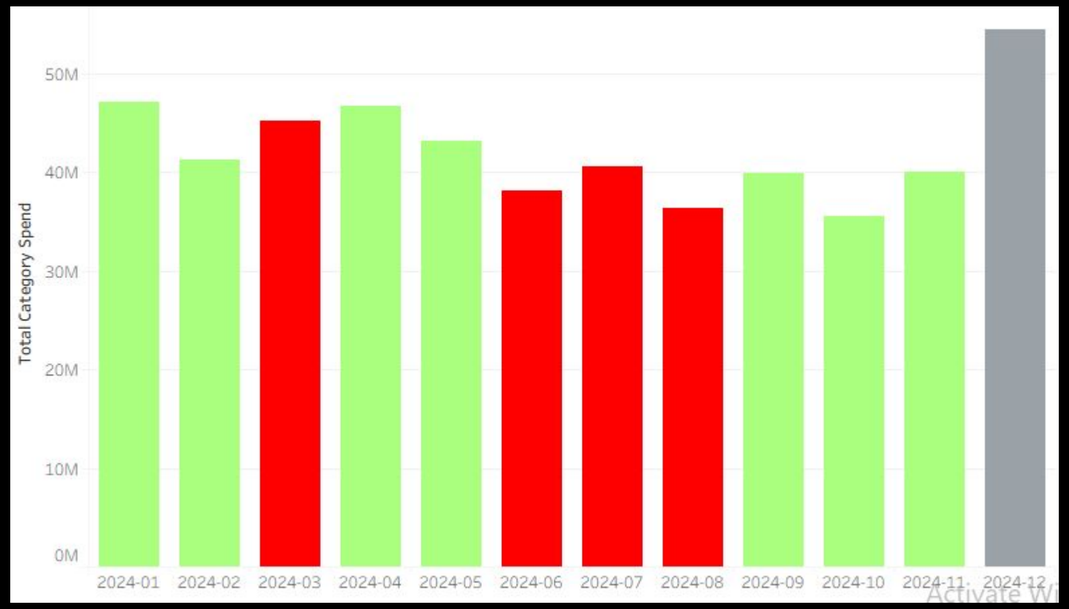
	Category									
Month of Date	Dining	Education	Entertai..	Groceries	Healthca..	Shopping	Subscrip..	Transport	Travel	Utilities
January 2024	8,01%	8,30%	13,96%	7,61%	11,99%	8,22%	7,20%	4,16%	9,01%	10,08%
February 2024	3,61%	10,97%	6,94%	6,17%	7,94%	10,38%	12,30%	5,70%	13,22%	4,05%
March 2024	10,42%	7,96%	3,59%	6,49%	5,40%	5,83%	6,28%	8,74%	12,50%	13,24%
April 2024	7,12%	15,50%	9,22%	8,67%	8,54%	7,82%	10,58%	6,93%	10,82%	5,33%
May 2024	4,85%	8,57%	8,66%	10,87%	5,69%	5,59%	17,24%	7,39%	6,33%	6,67%
June 2024	15,05%	7,20%	6,64%	4,59%	7,51%	4,95%	2,41%	4,72%	10,09%	6,93%
July 2024	6,79%	7,81%	11,37%	5,97%	8,54%	11,53%	9,49%	6,81%	3,85%	4,97%
August 2024	4,48%	7,76%	6,09%	2,08%	9,08%	8,66%	9,22%	7,01%	8,18%	6,78%
September 2024	9,47%	9,82%	8,53%	9,88%	10,51%	2,67%	9,86%	5,11%	6,68%	3,77%
October 2024	3,92%	7,72%	9,12%	4,56%	6,01%	8,95%	6,62%	8,10%	6,76%	5,42%
November 2024	10,64%	12,06%	6,76%	3,57%	5,64%	5,18%	9,31%	4,45%	8,21%	9,78%
December 2024	15,63%	7,44%	20,23%	6,46%	4,05%	3,55%	10,59%	14,22%	15,47%	6,32%

Budget Implementation

Month, Year of Date	# of Purchase	Amount	Differenc e in Amo..	Budget Variance
January 2024	90	Rp47M		-Rp2M
February 2024	84	Rp41M	-Rp6M	Rp4M
March 2024	85	Rp45M	Rp4M	Rp0M
April 2024	101	Rp47M	Rp1M	-Rp1M
May 2024	87	Rp43M	-Rp3M	Rp2M
June 2024	70	Rp38M	-Rp5M	Rp7M
July 2024	78	Rp41M	Rp2M	Rp5M
August 2024	67	Rp36M	-Rp4M	Rp9M
September 2024	80	Rp40M	Rp4M	Rp5M
October 2024	73	Rp36M	-Rp4M	Rp10M
November 2024	82	Rp40M	Rp5M	Rp5M
December 2024	103	Rp54M	Rp14M	-Rp9M

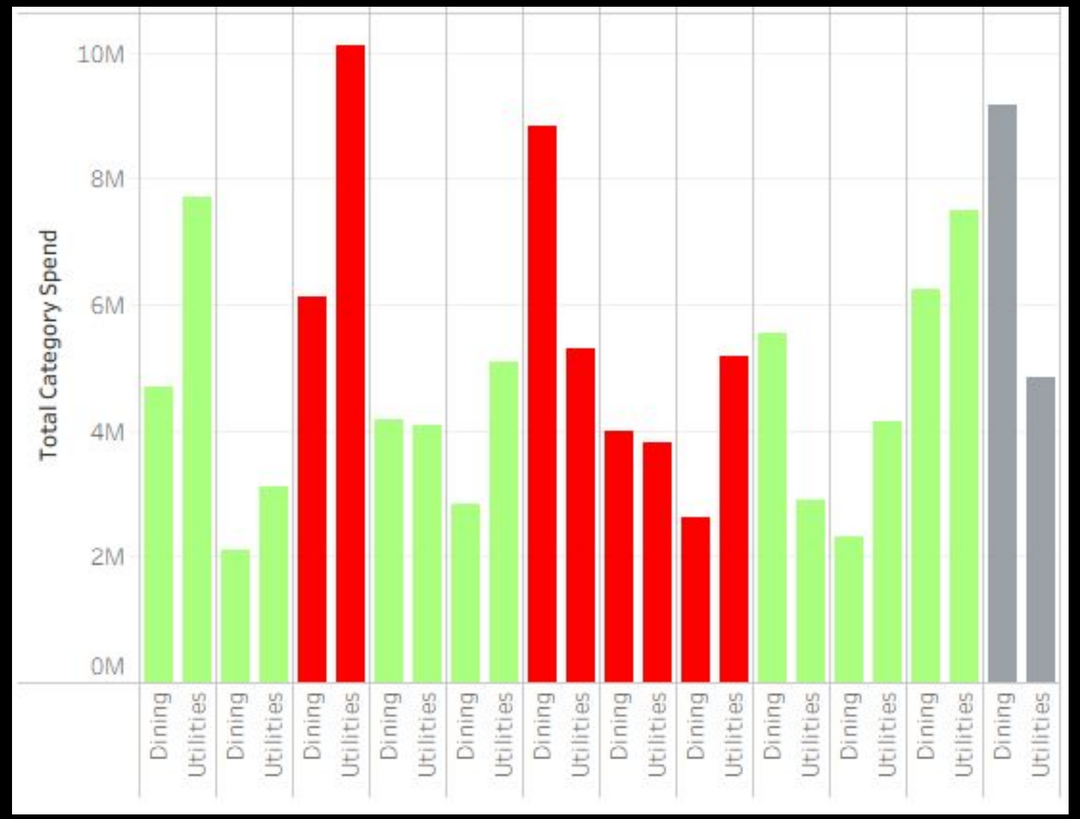
High Spend isn’t High Risk: Identifying Anomalous Behavior

- I make segmented users into 3 clusters: **Low, Emerging, and High Risk**.
- While **December** had the highest total spend, it remains **Low Risk** because the spending is predictable/seasonal.
- **Red Flag : There’s** a **4-month High-Risk window** in the middle of the year despite lower total spending.
- This indicates that **spending volatility and acceleration**, not absolute spend, drive risk.



Operational & Lifestyle Spikes Driving Portfolio Risk

- **March Outlier (Utilities):** Utility spending reached **Rp10M**, a sudden **300% increase** compared to February. This suggests unplanned operational overhead.
- **June Outlier (Dining):** Dining expenses surged to **Rp8.8M**, an **increase of over 200%** month-over-month. This indicates significant "Lifestyle Inflation" and impulsive discretionary spending.



RECOMMENDATIONS & NEXT STEPS

- Behavior-Based Budget Alerts : Trigger alerts when spending behavior shifts into a high-risk pattern, not only when budget is exceeded.
- Large Transaction Early Warnings : Notify users before confirming unusually large or fast-growing category spend.
- Category Spike Detection : Highlight sudden increases (e.g., Utilities, Dining) compared to previous months.
- Merchant & Location-Level Insights : Enable users to understand where and with whom most spending occurs.

Expected Impact

- Earlier intervention
- Reduced budget overruns
- Stronger user trust in RevoFinance as a financial coach, not just a tracker

OBIPR Strategic Framework

Observation	Business Impact	Isolation	Priority	Recommendation
• Total spending reached Rp509M, exceeding the planned budget. • Spending is highly concentrated across a small number of merchants and categories. • While December recorded the highest total spending, it remained within a Low Risk behavioral cluster. • In contrast, four mid-year months entered the High Risk cluster, driven by sudden category-level spending spikes rather than seasonal patterns.	Unmanaged behavioral spending volatility increases the risk of: • Budget overruns without early warning • Reduced user trust due to reactive (late) alerts • Missed opportunities for proactive financial guidance	Risk is isolated to specific categories (Utilities/Dining) during non-holiday months, suggesting these are not seasonal habits but behavioral anomalies.	• High - Behavioral risk alerts • High - Category spike detection • Medium - Large transaction warnings • Low–Medium Location-based insights	• Behavioral Risk Alerts: Trigger alerts when users move into a High Risk cluster based on spending acceleration, not just budget thresholds. • Category Spike Monitoring: Highlight unusual month-over-month increases in key categories such as Utilities and Dining. • Large Transaction Guardrails: Provide pre-transaction warnings for unusually large expenses relative to recent behavior. • Targeted Merchant & Location Insights: Focus controls and insights on top merchants and high-impact offline locations.

Thank You

https://www.linkedin.com/in/fitriyatus-sho/

Credits:

- Template Presentation by : Slide Cardinal